

Derek Thomas

AI SOLUTIONS ARCHITECT · ML SUCCESS ENGINEER

Abu Dhabi, UAE

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Profile Summary

AI Solutions Architect with 10+ years driving production AI adoption. Expert in Generative AI, LLM Deployment & Inference, Technical Communication, and Applied Engineering. Excels at the intersection of research, engineering, and customer enablement to deliver measurable business outcomes.

Experience

AI71 (Agentic AI Platform)

Abu Dhabi, UAE

PRINCIPAL SOLUTIONS ARCHITECT

Aug. 2025 – Present

- Embedded technical thought partner to internal AI engineering teams on system design, deployment architecture, and GenAI best practices
- Guided teams on production GenAI decisions: model selection in a fast-moving landscape, token reduction, and KV-cache optimization

Vendor Management

- Owned AI infrastructure vendor evaluation across four domains: OCR, VectorDB, WebSearch, and AutoTrain — built custom benchmarks, narrowed competitive landscapes, and drove final selection against multi-faceted requirements
- Led VectorDB benchmarking under demanding on-prem constraints: 5B+ documents, limited RAM — selected solution within strict infrastructure boundaries
- Identified and flagged potential fraud during AutoTrain vendor due diligence, protecting the company from a harmful contract

Customers

- Navigated tricky on-prem deployments to regulated government entities: communicated platform requirements and negotiated configuration boundaries
- Developed platform hardware calculator to accelerate scoping and estimation for new customers

Hugging Face (World-leading Open-source AI Unicorn)

Abu Dhabi, UAE

MACHINE LEARNING SUCCESS ENGINEER

Dec. 2022 – Mar. 2025

- Pre- and post-sales engineer, winning new business and enabling enterprise customers to create business value with GenAI.

Highlights

- Drove Hugging Face's largest-ever quarter (Q1 2024): single-handedly enabled Account Executives to close 120% of team quota
- Advised teams on cost-efficient Foundation Model strategies: Inference Optimization, Quantization, and Compact Model adoption
- Publications & Research: authored technical blogs on Arabic RAG, Speech-to-Speech, Multi-LoRA Inference, and more

GenAI Customer Guidance

- Guided teams (startups to Fortune 500): Agentic Workflows, prompt engineering, synthetic data generation, and LLM evaluation for Finance & Healthcare
- Fine-tuned LLMs for document summarization, code-gen, automated reporting, and domain-specific Agentic RAG (PyTorch/Hugging Face)
- Profiled and optimized large-scale inference pipelines at billion-classification scale

Platform Products

- HUGS — championed and deployed production-ready, zero-config inference microservices to accelerate customer time-to-value
- SafeCoder — drove adoption of self-hosted, compliant code-gen solution fine-tuned on customer private code

Leadership & Communication

- Thought Leadership: speaker at GITEX Dubai and leading AI summits in Seoul and Singapore
- Cross-Functional Communication: bridged technical and non-technical discussions with C-suite, engineering, and research teams
- Team Growth & Leadership: spearheaded recruitment, growing the ML Success team by 300%

DataRobot (AutoML Platform)

Abu Dhabi, UAE

CUSTOMER FACING DATA SCIENTIST

Nov. 2021 – Sep. 2022

- First Data Scientist in the Middle East; tailored Customer Success strategies to local needs distinct from established US & Europe frameworks

DataRobot-Hugging Face Compatibility

- Conducted largest-ever HF partner webinar in DataRobot history
- Pioneered AutoML and MLOps monitoring with Transformers (text drift, concept drift)
- Led DataRobot's first client enablement session with Transformers

End-to-End AI Solutions

- Delivered 20+ use cases from framing to production across 10 clients
- Developed specialized training and provided on-going Data Science up-skilling
- Empathized with customers to build champions at multiple levels

Industry Impact

- Banking: Propensity models & Next Best Offer boosted conversion rate 5x–13x per product
- Manufacturing: Machine efficiency & Material Quality drove \$1.6M in savings/yr
- Green AI: reduced CO₂ emissions nearly 200,000 tons per year
- Utilities: Solar Generation, Materials Demand Forecasting, Payment Default, et al.

UAE Government Contractor

LEAD DATA SCIENTIST / LEAD DATA ENGINEER

- Built hybrid cloud Big Data platform processing 6M documents/day as team lead
- Replaced legacy NLP/Geoparsing vendor solutions with Transformers to save \$1.3M
- Improved 20% recall (sentiment); 23% accuracy (NER); averaged across languages
- Managed data vendor relations and created an engineering solution to save \$600,000

Abu Dhabi, UAE
Feb. 2021 – Sep. 2021

G42 (AI Startup) → Unicorn)

SENIOR DATA SCIENTIST

Abu Dhabi, UAE
Apr. 2019 – Feb. 2021

Machine Translation — Arabic ↔ English

- Performance competitive with Google (± 3 BLEU points depending on domain)
- Led Data Engineering/Cleaning responsible for 78% of BLEU improvement from initial model
- Created fast, parallel data cleaning pipeline with Apache Airflow: 24B sentences

News Information Extraction

- Dynamic knowledge graph inter-connecting streaming news data with 40K articles/day
- Implemented and improved SotA Entity Linking models to be compatible with BERT

Mixed Language OCR

- Developed adaptive localized binarization filter in PyTorch; improved bounding box recognition and decreased character error rate by 50%

Saal.ai (AI Startup)

DATA SCIENTIST

Abu Dhabi, UAE
Mar. 2018 – Apr. 2019

- **Oil Price Prediction** (7-day forecasting): cleaned and feature engineered diverse time-series data of varying frequencies
- Developed NLP/AI microservices (Dockerized REST): Stance Detection (BERT semantic distance) and Graph Attention Networks for Text Classification

Rockwell Collins (Government Avionics)

DATA ANALYST; SYSTEMS ENGINEER; FPGA ENGINEER

Cedar Rapids, Iowa, USA
Jan. 2014 – Dec. 2017

- Developed a real-time data model yielding over \$15,000 in savings its first week of deployment

Publications

- 2020 **CIKM**, Relation Extraction with Self-determined Graph Convolutional Networks
- 2020 **ACL**, Autoencoding Keyword Correlation Graph for Document Clustering
- 2020 **SIGIR**, Attending to Inter-sentential Features in Neural Text Classification

Education

University of Louisville

MENG ELECTRICAL & COMPUTER ENGINEERING — *Magna Cum Laude* + Honours, GPA **4.0**

Louisville, KY, USA
Dec. 2011 – Dec. 2012

University of Louisville

BSC ELECTRICAL & COMPUTER ENGINEERING, MATHEMATICS MINOR — GPA **3.6**

Louisville, KY, USA
Aug. 2007 – Dec. 2011